

CIRCLE THEOREMS

Higher | Proofs

Circle theorem proofs and geometric reasoning

Name

Date

9 questions

32 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

Give clear mathematical reasons. Follow any question-specific restrictions on which results you may use.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers 2017-2025. Questions are included only in the booklet that matches their main purpose.

Question index

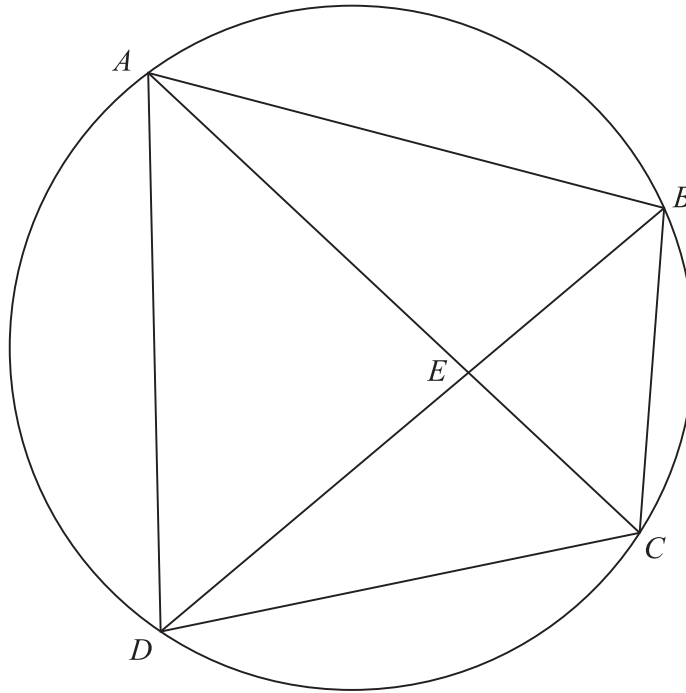
Higher | Proofs | 9 questions | 32 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	June 2017	1MA1/2H	Q15	3	3
02	November 2017	1MA1/3H	Q20	4	4
03	June 2018	1MA1/2H	Q13(a)-(b)	4	5
04	November 2018	1MA1/2H	Q21	4	6
05	November 2019	1MA1/1H	Q22	4	7
06	June 2022	1MA1/3H	Q15	3	8
07	November 2022	1MA1/1H	Q18	3	9
08	June 2024	1MA1/2H	Q22	4	10
09	November 2024	1MA1/2H	Q18	3	11

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

15 A , B , C and D are four points on the circumference of a circle.

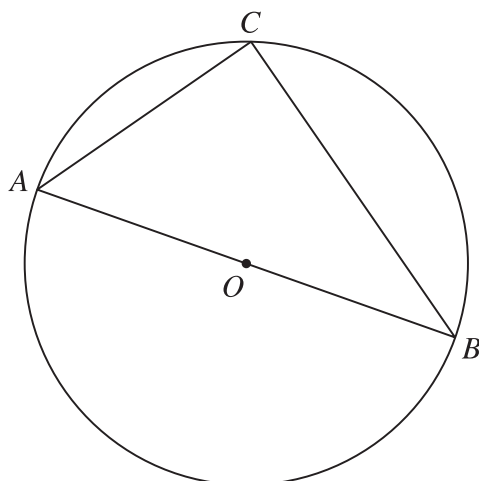


AEC and BED are straight lines.

Prove that triangle ABE and triangle DCE are similar.
You must give reasons for each stage of your working.

(Total for Question 15 is 3 marks)

20



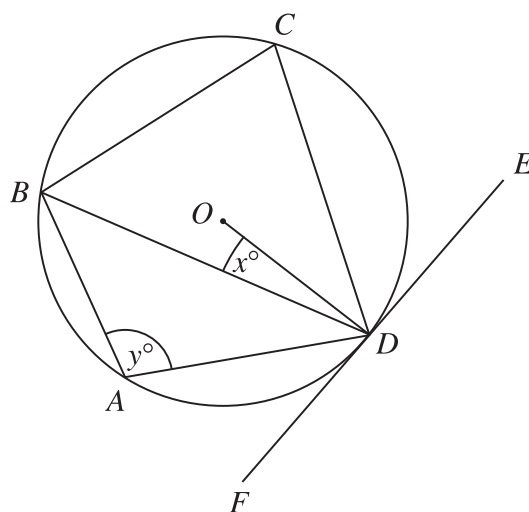
A , B and C are points on the circumference of a circle, centre O .
 AOB is a diameter of the circle.

Prove that angle ACB is 90°

You must **not** use any circle theorems in your proof.

(Total for Question 20 is 4 marks)

13



A , B , C and D are points on the circumference of a circle, centre O .
 FDE is a tangent to the circle.

- (a) Show that $y - x = 90$
 You must give a reason for each stage of your working.

(3)

Dylan was asked to give some possible values for x and y .

He said,

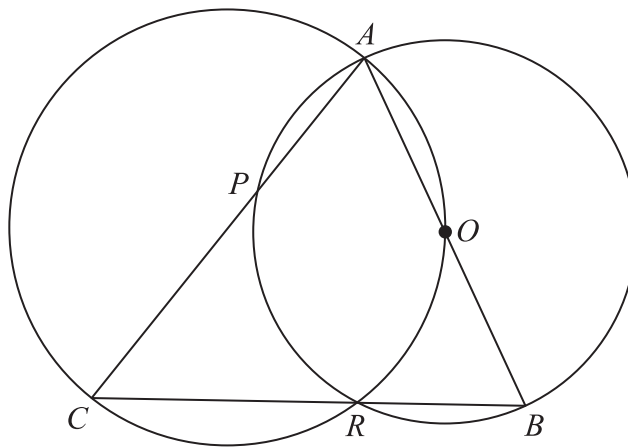
“ y could be 200 and x could be 110, because $200 - 110 = 90$ ”

- (b) Is Dylan correct?
 You must give a reason for your answer.

(1)

(Total for Question 13 is 4 marks)

21



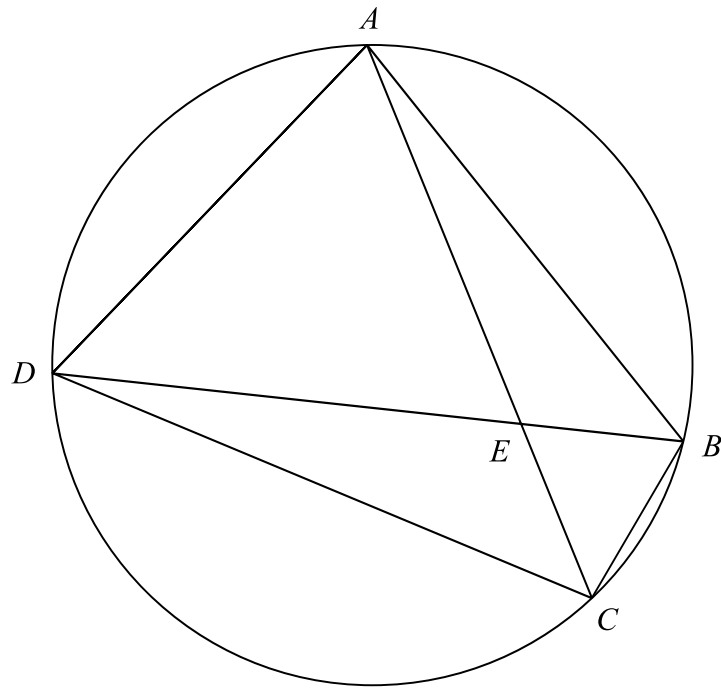
A , B , R and P are four points on a circle with centre O .
 A , O , R and C are four points on a different circle.
The two circles intersect at the points A and R .

CPA , CRB and AOB are straight lines.

Prove that $\text{angle } CAB = \text{angle } ABC$.

(Total for Question 21 is 4 marks)

22 A , B , C and D are four points on a circle.



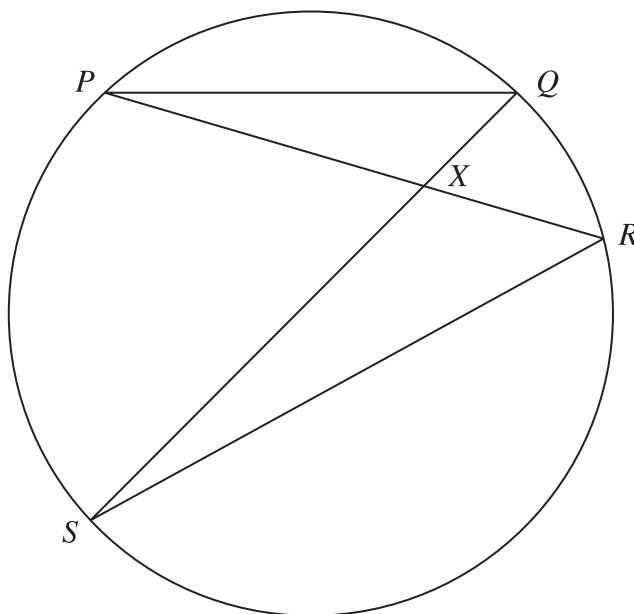
AEC and DEB are straight lines.

Triangle AED is an equilateral triangle.

Prove that triangle ABC is congruent to triangle DCB .

(Total for Question 22 is 4 marks)

15 P , Q , R and S are four points on a circle.

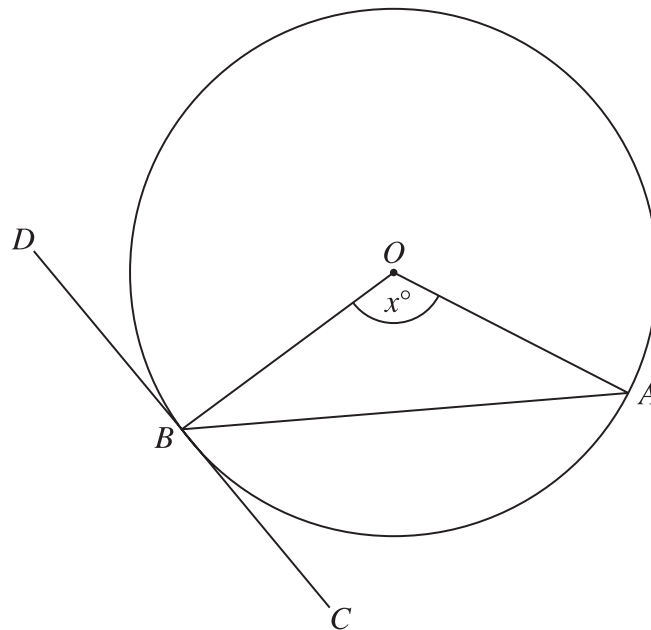


PXR and SXQ are straight lines.

Prove that triangle PQX and triangle SRX are similar.

(Total for Question 15 is 3 marks)

18



A and B are points on a circle, centre O .

DBC is the tangent to the circle at B .

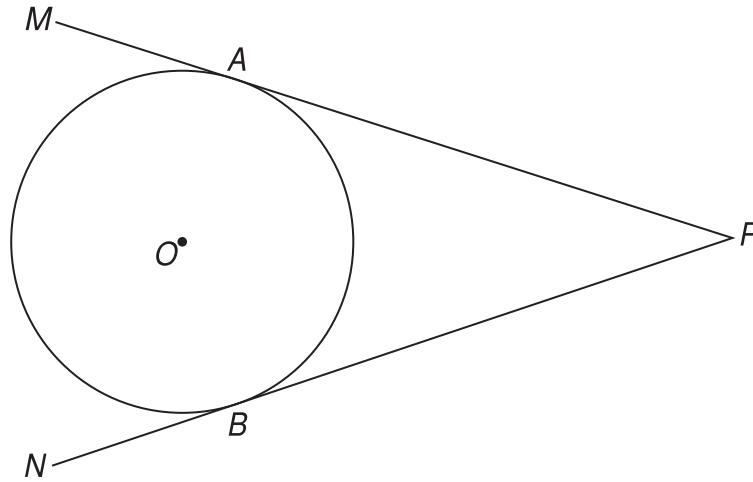
Angle $AOB = x^\circ$

Show that angle $ABC = \frac{1}{2}x^\circ$

You must give a reason for each stage of your working.

(Total for Question 18 is 3 marks)

22 A and B are points on a circle, centre O .

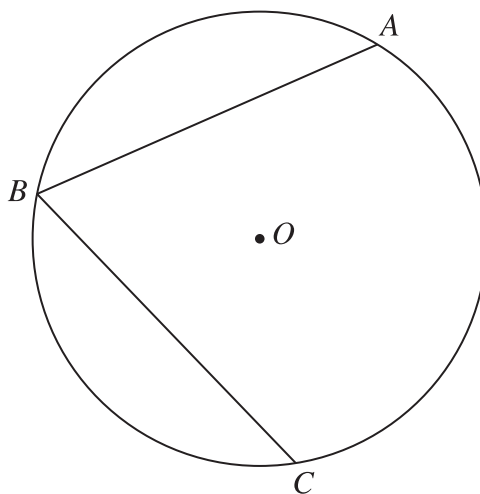


MAP and NBP are tangents to the circle.

Prove that $AP = BP$

(Total for Question 22 is 4 marks)

18 A , B and C are three points on a circle, centre O .



$$BA = BC$$

Prove that OB bisects angle ABC .

(Total for Question 18 is 3 marks)